

# Emotion Recognition from Facial Images

*Does Transfer Learning Improve Facial Emotion Classification Over a Baseline CNN?*

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## Abstract

We investigate whether ImageNet-pretrained transfer learning improves facial emotion classification over a custom convolutional neural network (CNN) trained from scratch on the FER-2013 benchmark. FER-2013 presents 35,887 low-resolution (48 x 48 grayscale) images across seven emotion categories with a severe class imbalance (16:1 ratio). Our experiments span four phases: a baseline CNN capacity study, a ResNet-50 transfer learning study with classical upscaling interpolation methods, a multi-model benchmark (VGG-16, ResNet-50, EfficientNet-B0), and systematic ablations of augmentation, loss functions, hyperparameters, and fine-tuning depth. A ResNet-50 trained on the full 28,709-image training set with a five-epoch frozen-backbone stage followed by fifteen epochs of full-network fine-tuning achieved a best weighted F1 of 0.696 and roughly 70% accuracy, compared with 0.598 weighted F1 and 61.1% accuracy for the strongest MS3 CNN. Phase 3 and the subsequent ablations use a smaller class-capped subset and are interpreted separately. We find that model architecture is the dominant performance driver within comparable experiments and that interpolation method has comparatively little effect within a fixed training setup. Grad-CAM visualizations show semantically plausible attention to facial regions, providing qualitative evidence of emotion-related feature learning.

## 1 Background and Motivation

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Automatic facial emotion recognition (FER) has wide-ranging applications in human-computer interaction, mental health screening, driver monitoring, and accessibility tools for individuals with autism spectrum conditions. Despite steady progress in deep learning, deployment in unconstrained environments remains difficult due to varying face angles, inconsistent lighting, partial occlusion, and low image resolution.

FER-2013, introduced by Goodfellow et al. (2013), is a standard benchmark of 35,887 low-resolution (48 x 48 grayscale) images collected via search engine queries - realistic but challenging. Our work addresses a practical question for any developer building an emotion-aware application: does the added complexity of transfer learning outperform a carefully tuned custom CNN, and what are the implications for computational budget and model interpretability?

A critical dimension of FER research is the intersection of class imbalance with demographic representation. FER-2013 was scraped without demographic controls, so underrepresented emotion classes (disgust at 1.5% versus happy at 25%) may map onto underrepresented racial or gender groups in the data. Failures on minority classes are therefore not neutral: if errors cluster by race or gender, the system operationalizes bias. Buolamwini and Gebru (2018) documented this pattern in commercial facial analysis systems, finding significantly higher error rates for darker-skinned individuals and women. This is why we prioritize macro F1 and per-class recall alongside accuracy throughout our experiments.

## 2 Problem Statement

**Research question:** Does ImageNet-pretrained transfer learning improve facial emotion classification performance over a baseline CNN trained from scratch on a low-resolution, class-imbalanced dataset?

Our target output is a seven-class emotion label drawn from angry, disgust, fear, happy, neutral, sad, and surprise. The unit of observation is a single 48 x 48 grayscale image. We refined our research question across milestones to emphasize robustness over raw accuracy: because class imbalance is severe and the dataset contains label noise, a model that achieves high overall accuracy by ignoring minority classes is not genuinely useful. We therefore report both weighted and macro F1 alongside accuracy, and we specifically track minority-class recall throughout.

## 3 Data and EDA

### 3.1 Dataset

FER-2013 contains 35,887 images split into 28,709 training and 7,178 test examples. Class counts are: happy (8,989), neutral (6,198), sad (6,077), angry (4,953), fear (5,121), surprise (4,002), and disgust (547). The 16:1 ratio between happy and disgust is the most extreme imbalance and strongly penalizes naive cross-entropy minimization.

| Emotion  | Train | Test  | Total | % of dataset |
|----------|-------|-------|-------|--------------|
| Angry    | 3,995 | 958   | 4,953 | 13.8%        |
| Disgust  | 436   | 111   | 547   | 1.5%         |
| Fear     | 4,097 | 1,024 | 5,121 | 14.3%        |
|          |       |       |       |              |
| Happy    | 7,215 | 1,774 | 8,989 | 25.0%        |
| Neutral  | 4,965 | 1,233 | 6,198 | 17.3%        |
| Sad      | 4,830 | 1,247 | 6,077 | 16.9%        |
| Surprise | 3,171 | 831   | 4,002 | 11.2%        |

Table 1: FER-2013 class distribution. Disgust is severely underrepresented at only 1.5% of the dataset.

### 3.2 EDA Findings

Three EDA analyses materially influenced our modeling decisions. First, per-class distributions of mean pixel intensity and pixel standard deviation are nearly indistinguishable across all seven emotion categories (Figure 1). This confirmed that raw intensity statistics carry little discriminative signal and that models relying on global pixel features would fail, motivating spatial feature learning via convolution.

Second, mean FFT magnitude spectra across emotion classes are nearly identical in structure, all dominated by low-frequency components. Emotions are essentially indistinguishable in frequency space. This reinforced the need for architectures that capture spatial geometry (eye shape, mouth curvature, brow position) rather than texture frequency.

Third, visual inspection of the training set revealed a meaningful number of noisy samples: duplicate groups of up to 14 near-identical images across different emotion labels, placeholder images (warning triangles and missing-image icons), and images of non-face objects. Although for MS4 we reverted to the original unmodified FER-2013 training

split to ensure reproducibility, these quality issues motivated our use of data augmentation and robust loss functions rather than assuming clean labels.

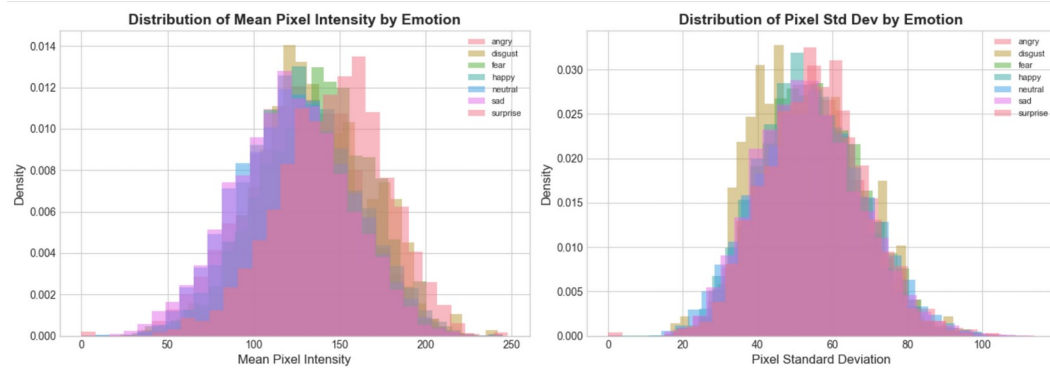


Figure 1: FER-2013 distributions of mean pixel intensity and pixel standard deviation.

## 4 Methods and Models

### 4.1 Preprocessing Pipeline

All images are loaded as RGB (three-channel) and normalized using ImageNet statistics (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]) for transfer-learning models, or converted to grayscale and normalized to mean 0.5 / std 0.5 for the Phase 1 BaselineCNN. Because ImageNet-pretrained models expect 224 x 224 inputs, the transfer-learning pipeline upscales images from 48 x 48 using four classical interpolation methods: Nearest Neighbor, Bilinear, Bicubic, and Lanczos. We use the FER-2013 canonical train/test split without carving out a separate validation set. Accordingly, metrics labeled validation in the notebook are actually metrics on the canonical test split. Because that split was consulted repeatedly across experiments, the results should be interpreted as exploratory comparisons rather than an unbiased final test estimate.

Phases 1 and 2 train on the full 28,709-image training set. To make the 16-run Phase 3 benchmark and subsequent ablations tractable, those experiments train on a class-capped subset of 3,436 images: up to 500 images per class, with all 436 available disgust images retained. Results are only directly comparable within the same training set and experimental protocol.

In the Part 2 transfer-learning ablation pipeline, baseline augmentation uses random horizontal flipping ( $p=0.5$ ), random rotation ( $\pm 15$  degrees), and color jitter (brightness  $\pm 20\%$ , contrast  $\pm 20\%$ ). We then compare None, Baseline, +Scaling, +Translation, +RandomCrop, and +RandomErasing. The separate Part 1 MS3 CNN-ablation pipeline uses  $\pm 10$ -degree rotation together with scaling, translation, random cropping, color jitter, and random erasing.

### 4.2 Baseline CNN Architecture

The MS3 capacity study evaluates four CNN architectures of increasing width and depth. All share a Conv2d  $\rightarrow$  BatchNorm2d  $\rightarrow$  ReLU  $\rightarrow$  MaxPool2d block pattern, while differing in channel widths, number of convolutional stages, fully connected depth, and dropout strength. Run1 (small, 2 stages, 32-64 channels) serves as an underfitting check; Run2 (baseline, 4 stages, 32-64-128-128) is the main reference; Run3 (deep, 4 stages, 32-64-128-256) increases abstraction; and Run4 (high-capacity, 4 stages, 64-128-256-512) is an overfitting stress test. These Run1-Run4 models are trained from scratch on the original 48 x 48 grayscale images.

A separate three-block BaselineCNN is used in the Phase 1 interpolation comparison. It operates on grayscale images upscaled to 224 x 224 so that interpolation methods can be evaluated under the same input-resolution pipeline as the pretrained models. Its results are not interchangeable with the stronger MS3 Run1-Run4 capacity-study results.

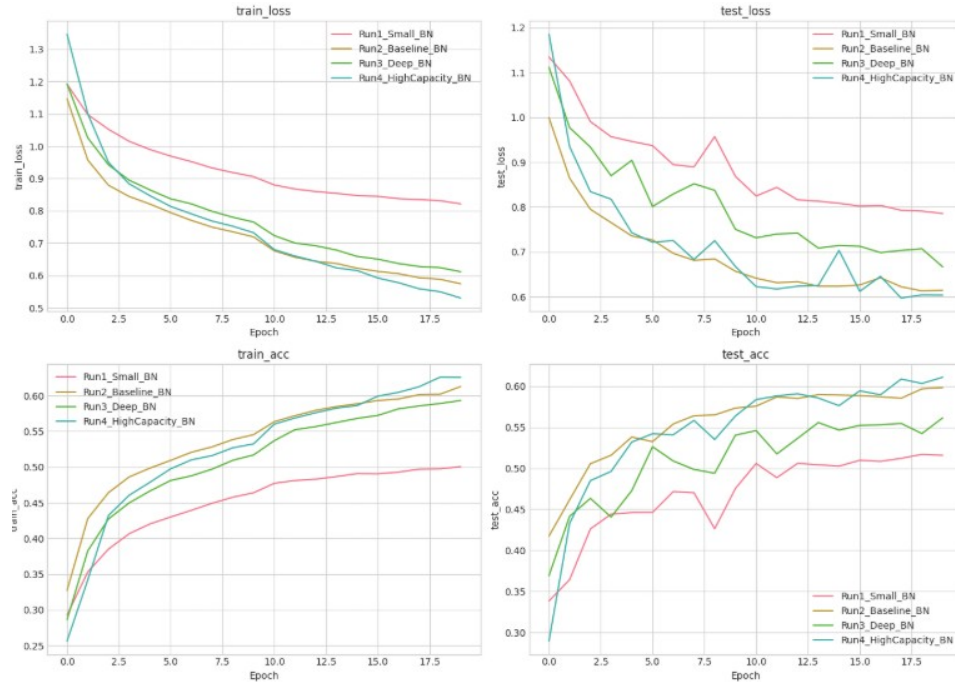


Figure 2: Training and test-split evaluation curves for the MS3 baseline CNN architectures with batch normalization.

### 4.3 Transfer Learning Models

We evaluate three pretrained models from torchvision: ResNet-50 (25.6M parameters), VGG-16 (138M parameters), and EfficientNet-B0 (5.3M parameters), all pretrained on ImageNet-1K. We replace the final classification head with a seven-class linear layer. Training follows a two-stage protocol. Stage 1 freezes the backbone for five epochs and trains only the head using Adam at  $\text{lr}=1\text{e-}3$ . Stage 2 unfreezes the full network for fifteen epochs and uses differential learning rates:  $\text{lr}=1\text{e-}4$  for the pretrained backbone and  $\text{lr}=5\text{e-}4$  for the classification head, with weight decay  $1\text{e-}4$  and cosine annealing.

### 4.4 Class Imbalance Strategies

We test five strategies on ResNet-50: Standard Cross-Entropy, Weighted Cross-Entropy, Focal Loss ( $\text{gamma}=2$ ), Weighted Random Sampling (WRS) + Standard CE, and WRS + Weighted CE. In the Phase 3 and subsequent subset-based experiments, class weights are computed in sklearn balanced mode from the 3,436-image class-capped subset; the corresponding samplers are also constructed from that subset.

### 4.5 Fine-Tuning Depth Study

Using the 3,436-image class-capped subset, we vary how much of the ResNet-50 and VGG-16 backbone is unfrozen: Head Only, Last Block + Head, Last 2 Blocks + Head, and an intended Full Fine-Tune condition. All strategies were trained for 20 epochs with cosine learning-rate annealing. A later code review found that the intended full-fine-tune condition did not correctly unfreeze the backbone; those results are invalid and are excluded from interpretation.

### 4.6 Grad-CAM Interpretability

We apply Gradient-weighted Class Activation Mapping (Grad-CAM, Selvaraju et al. 2017) to both the best CNN and the best ResNet-50. Grad-CAM computes the gradient of the class score with respect to the final convolutional feature map, produces a per-spatial-location importance weight, and overlays a heatmap on the original image. This provides a qualitative check on whether models are attending to facial geometry or confounding background patterns.

## 5 Results and Interpretation

### 5.1 CNN Capacity Study (MS3 Baseline)

All four MS3 CNN architectures were trained on 48 x 48 grayscale images using Focal Loss and Adam (lr=1e-3, wd=1e-4). Results are shown in Table 2. Increasing capacity consistently improved performance up to Run4, but the gap between accuracy and macro F1 across all runs reveals a persistent bias toward majority classes. In the best CNN configuration (Run4\_HighCapacity\_BN), the disgust class achieved near-zero recall, confirming that capacity alone cannot overcome severe class imbalance. The best MS3 CNN achieved 0.611 accuracy, 0.511 macro F1, and 0.598 weighted F1, while the MS3 frozen ResNet-50 reached only 0.445 accuracy and 0.356 macro F1. The best Phase 2 ResNet-50, trained on the full training set with a frozen-backbone warm-up followed by full-network fine-tuning, achieved 0.695 accuracy and 0.696 weighted F1. This is an 8.4 percentage-point accuracy gain and a 0.098 absolute weighted-F1 gain over the strongest MS3 CNN.

| Model                | Accuracy | Macro F1 | Weighted F1 | Precision (M) | Recall (M) |
|----------------------|----------|----------|-------------|---------------|------------|
| Run1_Small_BN        | 0.516    | 0.429    | 0.512       | 0.424         | 0.436      |
| Run2_Baseline_BN     | 0.598    | 0.489    | 0.579       | 0.492         | 0.499      |
| Run3_Deep_BN         | 0.561    | 0.463    | 0.551       | 0.522         | 0.457      |
| Run4_HighCapacity_BN | 0.611    | 0.511    | 0.598       | 0.644         | 0.520      |

Table 2: CNN capacity study results (MS3 baseline). Run4 performs best, but the macro-F1 gap reveals class-imbalance bias.

## 5.2 CNN Ablation Study

We applied the Run4 high-capacity architecture to a systematic ablation varying loss function, optimizer, learning rate, and weight decay. The Weighted Sampler + standard CE configuration achieved the best macro F1 of 0.535 and best macro recall of 0.567, providing more balanced gradient updates. Standard cross-entropy baseline accuracy was 0.592, Weighted CE reached 0.587, and Weighted Sampler CE reached 0.582 accuracy but 0.535 macro F1. Lower learning rates (1e-5) substantially degraded performance, and SGD was consistently inferior to Adam in these runs. Happy (F1 approximately 0.80-0.83) and surprise (approximately 0.70-0.74) were detected most reliably, while disgust (0.00-0.27) and fear (0.15-0.35) remained the hardest classes.

## 5.3 Transfer Learning: Interpolation and Architecture

In the full-data Phase 1 and Phase 2 comparison, the separate 224 x 224 BaselineCNN achieved 0.194 weighted F1, while ResNet-50 reached 0.696 weighted F1. This is a 3.6x improvement within that phase-specific comparison. Because the stronger MS3 high-capacity CNN achieved 0.598 weighted F1 under a different architecture and setup, we report that comparison separately rather than describing the Phase 1 model as the best from-scratch CNN.

| Phase                  | Model       | Upscaling  | Training data   | Test Acc. | Weighted F1 |
|------------------------|-------------|------------|-----------------|-----------|-------------|
| Phase 1 - Baseline CNN | BaselineCNN | Nearest N. | 28,709 full set | ~0.31     | 0.194       |
| Phase 2 - ResNet-50 TL | ResNet-50   | Lanczos    | 28,709 full set | ~0.70     | 0.696       |
| Phase 3 - Multi-Model  | VGG-16      | Lanczos    | 3,436 subset    | 0.5775    | 0.5795      |

| Phase                 | Model           | Upscaling | Training data | Test Acc. | Weighted F1 |
|-----------------------|-----------------|-----------|---------------|-----------|-------------|
| Phase 3 - Multi-Model | ResNet-50       | Bilinear  | 3,436 subset  | ~0.58     | 0.580       |
| Phase 3 - Multi-Model | EfficientNet-B0 | Bilinear  | 3,436 subset  | ~0.55     | 0.553       |
| Phase 3 - Multi-Model | BaselineCNN     | Bicubic   | 3,436 subset  | ~0.15     | 0.046       |

Table 3: Best results by phase and model. Phases 1-2 use the full training set; Phase 3 uses the 3,436-image class-capped subset. Cross-phase differences should not be attributed to upscaling alone.

Within the class-capped Phase 3 benchmark, interpolation had limited impact on transfer-learning performance: across the four upscaling methods, ResNet-50 weighted F1 ranged from 0.5678 to 0.5804, an absolute spread of 0.0126. Differences for VGG-16 and EfficientNet-B0 were similarly small. The larger difference between the Phase 2 ResNet-50 result (0.696) and the Phase 3 ResNet-50 result (0.5804) cannot be attributed to Lanczos versus Bilinear because Phase 2 used roughly eight times as much training data and a different experimental run. These results do not indicate a need for learned super-resolution preprocessing within a fixed experimental setup.

## 5.4 Augmentation Ablation

All augmentation-ablation results in this subsection use the 3,436-image class-capped subset. For ResNet-50, baseline augmentation (flip + rotation + color jitter) reached weighted F1 of 0.577 from a no-augmentation baseline of 0.544. Adding scaling raised this to 0.584. Translation, random crop, and random erasing produced 0.576, 0.574, and 0.579, respectively, showing diminishing returns. For the Phase 3 BaselineCNN, augmentation results remained low (0.000-0.169 weighted F1), indicating that augmentation could not compensate for insufficient representational capacity in that model.

## 5.5 Class Imbalance - Loss and Sampling (ResNet-50)

All loss-and-sampling results use the 3,436-image class-capped subset and subset-derived class weights. Weighted CE achieved the best weighted F1 at 0.592, narrowly outperforming standard CE (0.583), Focal Loss (0.581), WRS + Standard CE (0.576), and WRS + Weighted CE (0.577). The margins were small (within 0.016), suggesting that loss choice was secondary once the pretrained backbone was adapted in this subset-based experiment. Focal Loss did not emerge as a clear winner here, unlike the separate MS3 CNN ablation where Weighted Sampler provided the largest minority-class lift.

## 5.6 Hyperparameter Sensitivity (ResNet-50)

All hyperparameter-sensitivity results use the 3,436-image class-capped subset. Adam with the experiment's  $\text{lr}=1\text{e-}3$  setting outperformed SGD with momentum (0.578 versus 0.460 weighted F1). Reducing the configured learning rate to  $1\text{e-}4$  dropped weighted F1 to 0.544, and  $1\text{e-}5$  dropped it to 0.314. Weight decay had little effect (0.578 at  $\text{wd}=1\text{e-}4$ , 0.579 at  $\text{wd}=0$ , and 0.582 at  $\text{wd}=1\text{e-}3$ ). These findings are specific to the staged training implementation and subset used in this ablation.

## 5.7 Fine-Tuning Depth Study

All fine-tuning-depth results use the 3,436-image class-capped subset. Among the valid partial fine-tuning conditions, ResNet-50 performed best when the last two blocks plus the head were trainable, reaching 0.579 weighted F1. This exceeded head-only training (0.361) and last-block + head (0.535). VGG-16 showed a similar pattern: last two blocks + head reached 0.571, compared with 0.339 for head-only and 0.554 for last-block + head. The intended full-fine-tuning condition contained an unfreezing implementation error and cannot be interpreted; it must be rerun before any conclusion about full fine-tuning is made.



| Strategy                 | ResNet-50 F1 | VGG-16 F1 |
|--------------------------|--------------|-----------|
| Head Only                | 0.361        | 0.339     |
| Last Block + Head        | 0.535        | 0.554     |
| Last 2 Blocks + Head     | 0.579        | 0.571     |
| Full Fine-Tune (invalid) | Invalid      | Invalid   |

Table 4: Fine-tuning depth study on the 3,436-image subset. The intended full-fine-tune condition did not correctly unfreeze the backbone and is excluded from interpretation.

## 5.8 Grad-CAM Interpretation

Grad-CAM heatmaps from the best baseline CNN show reasonable but diffuse attention, often spreading across the central face region with some focus on the nose and mouth area. Correctly classified samples (angry, happy, neutral) showed tighter activation on diagnostic regions. Misclassified samples (disgust predicted as surprise, sad predicted as fear) showed the CNN attending to incorrect or peripheral regions.

ResNet-50 Grad-CAM heatmaps showed more spatially concentrated and semantically plausible attention. Happy predictions concentrated on the mouth region, angry predictions activated strongly on the brow-eye region, and surprise activations centered on widened eyes. These visualizations provide qualitative, not causal, evidence that the model relies on facial regions associated with the predicted emotions.

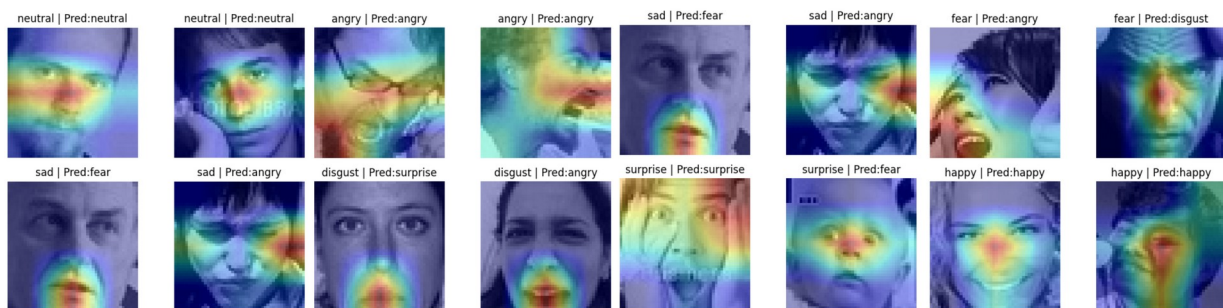


Figure 3: Baseline CNN Grad-CAM examples showing correct predictions and misclassifications (for example, disgust -> surprise and fear -> angry).

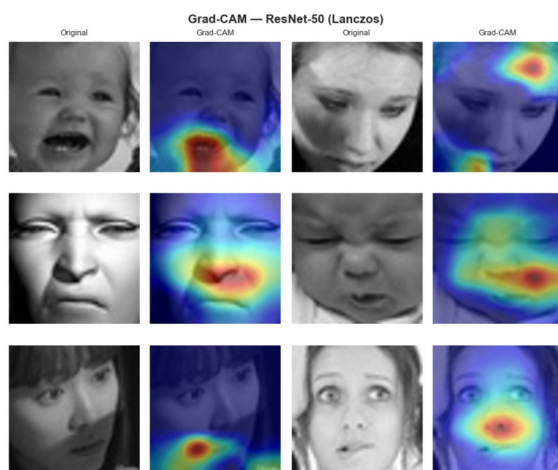


Figure 4: Grad-CAM heatmaps from ResNet-50 (Lanczos). Activations concentrate on plausible diagnostic regions, including the mouth for happy and the brow for angry.

## 6 Conclusions and Discussion

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ImageNet-pretrained transfer learning outperformed the custom CNN baselines on FER-2013. A Phase 2 ResNet-50 trained on the full 28,709-image training set with a frozen-backbone warm-up followed by full-network fine-tuning achieved 0.696 weighted F1 and 0.695 accuracy, versus 0.598 weighted F1 and 0.611 accuracy for the strongest MS3 CNN. This is a 0.098 absolute weighted-F1 gain and an 8.4 percentage-point accuracy gain. Grad-CAM provides qualitative evidence of more spatially concentrated attention.

Several secondary findings are worth emphasizing. First, interpolation method had limited impact within fixed-model, fixed-training-set comparisons; the exact best method varied by experiment. Second, the Phase 3 benchmark and all subsequent transfer-learning ablations used a 3,436-image class-capped subset, so their values should not be directly compared with the full-data Phase 2 result as though upscaling or model choice were the only differences. Third, among the valid subset-based partial fine-tuning runs, last two blocks + head performed best for both ResNet-50 and VGG-16. The intended full-fine-tuning condition contained an unfreezing implementation error, so no conclusion about full fine-tuning or catastrophic forgetting can be drawn until that experiment is rerun. Fourth, moderate augmentation captured most of the available benefit, while additional transforms showed diminishing returns.

What we cannot claim: our results are limited to a single benchmark under a fixed set of architectures. FER-2013 is widely considered to have noisy labels, and findings such as Adam outperforming SGD in our runs may not generalize across datasets with different noise profiles. We do not evaluate inference latency or model size, which matter for deployment on edge devices, and we did not include Vision Transformer architectures or newer efficient backbones. Because the canonical test split was consulted repeatedly across experiments, the reported scores are exploratory comparisons rather than an unbiased final test estimate. The most fundamental limitation is the supervision signal: FER-2013's hard one-hot labels poorly reflect the continuous, ambiguous nature of human emotion. Soft re-labeling from annotator-agreement distributions, or joint valence/arousal prediction alongside discrete classes, would better match this structure. A second priority is demographic robustness: Grad-CAM visualizations do not establish equal reliability across skin tones and genders. Fairness-aware training objectives and demographically stratified evaluation sets should be built in from the outset. Better supervision and fairer evaluation are more likely to produce meaningful real-world gains than architectural scaling alone.

## 7 Broader Impact

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Emotion recognition from facial images raises significant ethical considerations that any deployment must address. On the positive side, accurate FER can benefit people who have difficulty reading social cues, enable assistive applications, and improve driver-safety systems. These applications could reduce harm and improve quality of life.

However, the potential harms are substantial. Emotion inference from faces is not reliable across demographic groups; published studies have documented higher error rates for darker-skinned faces and for women in commercial facial-analysis systems. FER-2013 itself was collected via search-engine queries and may not represent demographic diversity. A model trained on this data could perpetuate or amplify disparities in high-stakes settings such as hiring, policing, or mental-health triage. FER systems could also be used for covert emotional surveillance without consent, with implications for privacy and civil liberties.

We recommend that any production deployment include: (1) demographic bias audits before release, (2) explicit consent from people being monitored, (3) human oversight for consequential decisions, and (4) opt-out mechanisms. Our work is a research study on a public benchmark and should not be interpreted as a production-ready system.



## References

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- [1] Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., Chumerin, N., & Bengio, Y. (2013). Challenges in representation learning: A report on three machine learning contests. ICML 2013 Workshop on Challenges in Representation Learning, arXiv:1307.0414.
- [2] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In Proceedings of CVPR.
- [3] Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. arXiv:1409.1556.
- [4] Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. In Proceedings of ICML.
- [5] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In Proceedings of ICCV.
- [6] Lin, T. Y., Goyal, P., Girshick, R., He, K., & Dollar, P. (2017). Focal loss for dense object detection. In Proceedings of ICCV.
- [7] FER-2013 Dataset. Available at: <https://www.kaggle.com/datasets/msambare/fer2013>
- [8] PyTorch / TorchVision. (2023). torchvision.models. <https://pytorch.org/vision/stable/models.html>
- [9] Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. In Proceedings of the 1st Conference on Fairness, Accountability and Transparency (FAT\* 2018), PMLR 81, pp. 77-91.
- [10] AI Assistance: Claude (Anthropic) was used for report editing/grammar checks. All experiments, analysis, code, and conclusions are the work of the authors.